

PROJECT PROPOSAL

🌐 “Global Lifestyle Impact Predictor (GLIP):

A Multi-Dimensional AI-Powered 3D System for Daily Habit Impact Analysis”

~carbon emission



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1. Introduction

1.1 Global Lifestyle Impact & Need for Prevention

Sustainability challenges such as climate change[19], water scarcity, plastic pollution[20], and e-waste generation[17] are growing rapidly. Daily human lifestyle choices — travel, electricity use, shopping, food consumption, and gadget usage — directly fuel these global threats. Individuals and organizations often underestimate these impacts because many existing tools present only isolated metrics rather than a holistic view across multiple environmental dimensions. The proposed system aims to close this awareness gap by providing a consolidated, scientifically grounded assessment of lifestyle impacts across multiple domains.

1.2 Importance of Sustainability in Modern Society

Millions of individuals and organizations lack understanding of their true environmental footprint[32][34]. Existing platforms typically focus only on carbon emissions[25][30] and ignore other critical environmental metrics such as water footprint, plastic waste[20][27], e-waste[17][29], and forest loss. This narrow framing reduces the effectiveness of awareness campaigns and policy interventions that require multi-dimensional evidence. GLIP's multi-output approach will enable users and institutions to see trade-offs and co-benefits (for example, food choices that affect both water use and forest loss[16]) so decision-making can be better informed and aligned with UN Sustainable Development Goals (SDGs)[21].

1.3 Limits of Traditional Carbon Calculators

Traditional calculators commonly:

- Provide only **single-output CO₂ values**[25][33].
- Contain **no prediction models** for future trajectories.
- Are **non-AI systems** (static rules / lookup tables).
- Lack **personalization & actionable recommendations**.
- Do not cover **multi-dimensional environmental impact** (water, plastic[20], e-waste[17], forest loss).

These limitations prevent meaningful, long-term behavior change because users see isolated numbers rather than context, forecasts, or tailored alternatives. GLIP is designed to directly address each of these shortfalls.

1.4 Role of Artificial Intelligence in Multi-Impact Analysis

AI enables the conversion of structured lifestyle inputs into multiple outputs simultaneously (e.g., water, plastic, e-waste, CO₂, forest loss). Specific roles of AI in GLIP:

- **Multi-output mapping:** map inputs from five categories (travel, food, shopping, gadgets/technology, electricity) into several environmental impact metrics[26].
- **Forecasting:** use LSTM/RNN and related time-series models to predict 5–10 year future impacts if current habits continue[25].
- **Personalized recommendations:** recommend realistic alternatives (e.g., dietary swaps, transport choices) that balance user constraints with environmental gains.
- **Pattern discovery:** detect user clusters and behaviour patterns to generate community/organization level insights and targeted campaigns[26][31].

1.5 Objective of the Proposed Framework

- Build the **first multi-output lifestyle impact prediction system** covering CO₂, water, plastic, e-waste, and forest loss.
- **Integrate AI + 3D visualization [11][12]** (interactive globe) to show immediate and projected impacts.
- **Provide predictions + personalized recommendations** that are practical and data-backed.
- **Promote global sustainability** through an engaging, research-driven tool that supports individuals, NGOs, and policy makers.

2. Literature Review

A key related eight research papers:

Research papers whose limitations are fulfilled by our project GLIP(Global Lifestyle Impact Predictor)

- **“Application of Artificial Intelligence to Predict CO₂ Emissions” (2023)**

Link: <https://www.mdpi.com/2071-1050/15/9/7648>

Reason for Selection

- Only predicts **carbon emissions**, not multi-dimensional impact.

- Uses simple ML → no long-term deep-learning predictions.
- No analysis of water footprint, e-waste, plastic waste, or forest loss.
- No interactive visualization or recommendations.

GLIP overcomes all these gaps.

▪ **“Machine Learning Methods to Study Environmental Footprint” (2020)**

Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7579113/>

Reason for Selection

- Covers only *partial* lifestyle data.
- Does not combine multiple categories (travel, food, shopping, gadgets, electricity).
- No user-specific dashboard or footprint prediction.
- No recommendation system

GLIP covers all lifestyle categories + prediction + personalized UX.

▪ **“Generation Patterns & Consumer Behavior of Single-Use Plastics (SUPs)” (2022)**

Link:

<https://www.sciencedirect.com/science/article/abs/pii/S0045653521035311>

Reason for Selection

- Focuses only on plastic waste.
- No integration with carbon, water, e-waste, or deforestation.
- Pure research — does not convert findings into a usable system.

GLIP integrates plastic + all other impact types in one system.

▪ **“Analyzing Behavioral Patterns in Plastic Waste Management” (2024)**

Link: <https://www.atlantispress.com/article/126003918.pdf>

Reason for Selection

- Only behavioral study; lacks AI or prediction.
- No multi-impact environmental calculation.
- No digital tool or interactive representation.

GLIP uses AI + 3D + multi-output analysis → huge improvement.

▪ **“E-Waste Footprint Estimation Models for Smart Device Users” (2018)**

Link: [https://shura.shu.ac.uk/25051/1/Mostafiz-ProposedEwasteManagement\(AM\).pdf](https://shura.shu.ac.uk/25051/1/Mostafiz-ProposedEwasteManagement(AM).pdf)

Reason for Selection

- Only computes e-waste impact.
- No combined calculation with carbon or plastic or water.
- No future predictions or recommendations.
- No 3D visualization or dashboards.

GLIP merges e-waste with four other impact types + AI forecasting.

- **“AI for Calculating and Predicting Building Carbon Emissions” (2025)**

Link: <https://link.springer.com/article/10.1007/s10311-024-01799-z>

Reason for Selection

- Predicts carbon emissions only.
- Limited to building sector — not user lifestyle.
- No multi-dimensional impact scoring.

GLIP is lifestyle-focused + multi-impact + user-personalized.

- **“Machine Learning–Based Life Cycle Assessment for Environmental Impact” (2024)**

Link: <https://setac.onlinelibrary.wiley.com/doi/10.1002/ieam.4954>

Reason for Selection

- Provides LCA but as research, not a usable system.
- No unified multi-output model.
- No 3D visualization or user experience design.

GLIP transforms research → real-world actionable tool.

- **“Lifestyle Carbon Footprints and Changes in Lifestyles to Limit Global Warming to 1.5 °C” (2021)**

Link: [Lifestyle carbon footprints and changes in lifestyles to limit global warming to 1.5 °C, and ways forward for related research | Sustainability Science](#)

Reason for Selection

- Focuses only on environmental factors (food, housing, mobility).
- No inclusion of health impact domain.
- No personalization or user-specific feedback.

GLIP integrates health and environmental factors for a complete lifestyle assessment.

GLIP provides personalized recommendations using real-time user data.

3. Problem Statement

People do not have a clear understanding of how their everyday lifestyle choices—such as travel, food consumption, shopping habits, gadget usage, and electricity consumption—collectively impact the environment[17][19]. Existing tools focus almost entirely on **carbon emissions**[25][30], ignoring other critical dimensions like **water usage**[16], **plastic generation**[20][27], **e-waste production**[17][19], **and forest loss**. Because current calculators are simple, non-AI, single-output systems with no predictions, no recommendations, and no interactive visualizations, users receive fragmented, incomplete, and non-actionable insights.

This lack of a **holistic, multi-dimensional, user-friendly, and predictive system** prevents individuals, organizations, and communities from understanding their true environmental footprint and making informed changes. Therefore, there is a need for an AI-powered platform that integrates **multi-output impact calculation, future forecasting, personalized sustainability recommendations, and engaging 3D visualization** to create awareness, motivate behavior change, and support global sustainability goals.

4. Proposed Methodology

4.1 Operational Framework of GLIP

The operational workflow of the Global Lifestyle Impact Predictor (GLIP) consists of multiple stages that together form a complete multi-impact assessment and visualization system.

▪ User Input Collection

Users enter their lifestyle information across five major categories:

- Travel
- Food
- Shopping
- Gadgets & Technology
- Electricity Consumption

All inputs are structured and standardized to ensure accurate processing.

▪ **Data Preprocessing**

Before analysis, the input data undergoes a preprocessing pipeline that includes:

- Cleaning and validation of user entries
- Normalization of values to maintain consistent scales
- Extraction of relevant features required for multi-impact calculations

This ensures reliability, consistency, and readiness for downstream computation.

▪ **Multi-Impact Calculation Engine**

The system converts the processed lifestyle data into measurable environmental indicators.

GLIP generates **five core outputs**:

- Carbon emissions
- Water footprint
- Plastic waste generation
- E-waste footprint
- Forest loss indicators

This multi-output design provides a holistic understanding of lifestyle impacts.

▪ **Prediction and Trend Estimation**

The system analyzes historical and user-input patterns to estimate **future environmental impact trends**.

The prediction module provides:

- Short-term and long-term footprint projections
- Habit-based scenario comparisons
- Possible future outcomes if current behaviors continue

The aim is to make users aware of long-term consequences, not just present-day impact.

▪ Recommendation Engine

Based on the calculated impacts and projected trends, GLIP generates **personalized sustainability recommendations**, such as:

- Alternative transport options
- Environmentally friendly shopping behaviors
- Reduced-impact dietary choices
- Better gadget usage and disposal habits
- Electricity conservation suggestions

Recommendations are designed to be realistic, practical, and user-specific.

▪ Visualization Module (3D + Dashboard)

The system presents results through an engaging visual interface, including:

- A **3D interactive globe** showing environmental footprints
- Category-wise impact comparison dashboards
- Visual symbols for CO₂, water usage, plastic, e-waste, and forest loss

This transforms numerical data into an intuitive and impactful visual experience, improving user understanding and engagement.

4.2 System Flow Diagram

The overall workflow of the proposed system is shown below. As illustrated in **Figure 1**, the proposed system processes user input through multiple stages, including preprocessing, impact calculation, prediction, recommendation, and 3D visualization. The process begins with collecting raw user input, which is then cleaned and structured during the data preprocessing stage. The **Multi-Impact Calculation Engine** analyzes multiple parameters to determine key performance indicators. The **Prediction and Trend Estimation** module forecasts future patterns based on historical and current data. Next, the **Recommendation Engine** suggests optimal solutions or actions tailored to the user's needs. Finally, the **3D Visualization Module** presents results interactively, making complex data easy to understand and analyze.

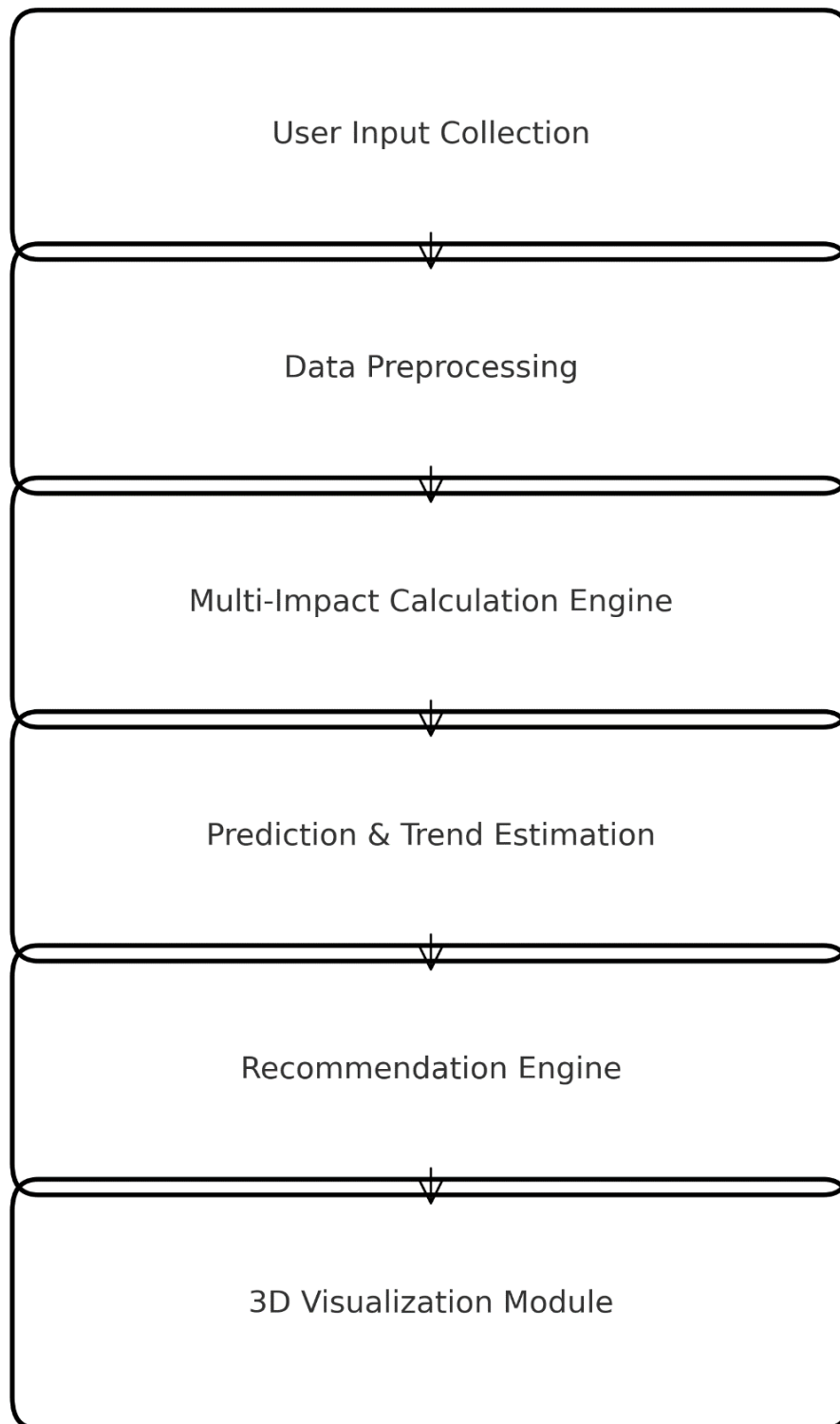


Figure 1: System Architecture showing sequential workflow from user input to 3D visualization.

5. Project Objectives

The **Global Lifestyle Impact Predictor (GLIP)** aims to build an advanced AI-powered web system that enables individuals and organizations to analyze, predict, and visualize the environmental consequences of their daily lifestyle choices. The project focuses on bridging scientific accuracy with accessibility through multi-dimensional impact analysis, AI-driven prediction, and engaging 3D visualization.

▪ Provide Multi-Dimensional Environmental Impact Analysis

- Assess user habits across **five lifestyle categories**: Travel, Food, Shopping, Gadgets/Technology, and Electricity.
- Convert lifestyle inputs into **multiple measurable outputs**, including carbon emissions, water footprint, plastic waste, e-waste footprint, and deforestation rate.
- Deliver a **holistic, science-based view** of how everyday actions contribute to global environmental change.
- Move beyond traditional carbon calculators by integrating **multi-faceted sustainability parameters**.

▪ Predict Long-Term Sustainability Impact

- Utilize **AI/ML-based models (LSTM/RNN)**[4][13][14][30][31] to forecast long-term environmental trends.
- Provide users with **5-year projections** of their carbon, water, plastic, and waste footprints if habits continue unchanged.
- Enable individuals and organizations to **anticipate future sustainability challenges** and plan eco-friendly interventions.
- Support data-driven awareness campaigns and policy recommendations using **predictive analytics**.

▪ Deliver Personalized Recommendations

- Offer **AI-driven, user-specific suggestions** to promote sustainable alternatives for every lifestyle area.
- Example recommendations include: switching from car to public transport, beef to lentils, or fast fashion to sustainable clothing brands.
- Ensure all suggestions are **practical, achievable, and personalized** to maintain motivation and behavior change.
- Help users reduce their environmental impact while maintaining comfort and convenience.

▪ Offer Visually Engaging 3D Representation

- Integrate **3D visualization technologies (Three.js, WebGL, Blender)[11][12]** to present environmental data in an interactive, immersive form.
- Represent each environmental dimension visually:
 - **CO₂ clouds** – Travel & Electricity
 - **Drying rivers** – Water consumption from Food
 - **Plastic in oceans** – Shopping-related waste
 - **Shrinking forests** – Food & Shopping impacts
 - **E-waste piles** – Gadget & Tech usage
- Create a **user-friendly and inclusive interface** with accessibility features such as screen-reader compatibility and color-blind-friendly visuals.
- Make sustainability learning **engaging, visual, and gamified** for a lasting impact.

▪ Support SDG-Based Impact Mapping

- Align all calculated environmental impacts with the **United Nations Sustainable Development Goals (SDGs)[21]**:
 - **SDG 13** – Climate Action (Travel & Electricity)
 - **SDG 6** – Clean Water and Sanitation (Food)
 - **SDG 12** – Responsible Consumption & Production (Shopping & Tech)
 - **SDG 14** – Life Below Water (Plastic Waste from Shopping)
 - **SDG 15** – Life on Land (Deforestation from Food & Shopping)
- Provide **bulk analysis capabilities** for organizations, NGOs, and educational institutions to measure group-level impact.
- Position GLIP as a **global educational and awareness tool** for promoting sustainable living and supporting CSR initiatives.

▪ Introduce a Multi-Input → Multi-Output Environmental Model

Our system pioneers a **multi-input → multi-output AI framework** that distinguishes between **indirect** and **direct** environmental impacts to provide a truly holistic sustainability analysis.

- **Indirect impacts** refer to those **emissions or effects that occur as a consequence of an activity**, rather than from the activity itself — such as **carbon emissions** generated through energy use or transportation.

- **Direct impacts** represent **immediate, measurable outcomes** of lifestyle habits, including **water consumption, plastic waste generation, e-waste accumulation, and deforestation** resulting directly from choices in food, shopping, and technology.

6. Dataset

The project will utilize a combination of **internationally recognized, scientifically validated datasets** to ensure accurate, reliable, and standardized environmental impact analysis. Data will be sourced from multiple global organizations and merged into a unified framework for AI-based processing and prediction.

Primary Data Sources

- **Lifestyle Datasets** – Collected from **FAO (Food and Agriculture Organization)[16][33]**, **United Nations (UN)[17]**, and the **Global Footprint Network[18]** to cover consumption habits, food patterns, transportation behavior, and energy use statistics across regions.
- **Carbon Emission Datasets[19]** – Provide verified carbon output data associated with transportation, electricity consumption, and industrial activities, enabling accurate CO₂ footprint estimation.
- **Water Footprint Database[16]** – Contains global and regional data on water usage in agriculture, food production, and daily consumption patterns, used to assess indirect water impacts.
- **E-Waste Datasets[17][29]** – Include data from global e-waste monitoring studies and UN reports, detailing device production, disposal rates, and recycling inefficiencies.
- **Plastic Waste Lifecycle Reports[20][27][28]** – Offer detailed insights into the production, usage, recycling, and environmental persistence of plastics, supporting plastic impact estimation for shopping and consumer activities.

Data Integration & Standardization

All datasets will be **merged, cleaned, and standardized** to establish a consistent input structure compatible with GLIP's **multi-input → multi-output AI model**. Data normalization will ensure uniform units, comparable metrics, and balanced representation across all environmental dimensions (carbon, water, e-waste, plastic, deforestation).

This comprehensive and harmonized dataset foundation will enable **accurate calculations, AI-driven predictions, and SDG-based sustainability mapping** across diverse lifestyle categories.

7. Tools and Technologies

- **Frontend Development:** For creating a responsive, user-friendly, and visually appealing interface-**html,css**[5][6][7][8][9][10].
- **Backend Development:** To manage server-side logic, API communication, and data processing[1][3][15].
- **Database Management:** For secure storage and retrieval of user data, impact calculations, and predictions-**mongodb**[2].
- **AI and Machine Learning:** To analyze lifestyle data and generate intelligent predictions and recommendations-**python**[13][14].
- **3D Visualization:** For developing interactive and engaging graphical representations of environmental impacts-**three.js**[11][12].
- **Hosting and Deployment:** To ensure scalability, reliability, and global accessibility of the system-**vercel/heroku/aws**.
- **Version Control:** For maintaining code versions, enabling collaboration, and managing updates efficiently-**git/github**.

8. Functional Requirements

8.1 User Authentication and Access

- The system shall allow new users to **sign up** using an email and password.
- The system shall allow existing users to **log in** securely using authentication mechanisms.
- The system shall maintain **session management** to prevent unauthorized access.
- The system shall provide **role-based access** (e.g., user, admin, organization).
- The system shall allow users to **log out** and automatically expire Inactive sessions.

8.2 Input Upload & Preprocessing

- The system shall allow users to **enter lifestyle information** across five main categories:
 - Travel
 - Food
 - Shopping

- Gadgets/Technology
- Electricity
- The system shall provide **structured and predefined input options** (drop-downs, sliders, or checkboxes).
- The system shall **validate and preprocess inputs** to ensure consistency and completeness.

8.3 Multi-Impact Detection (AI/ML Functionality)

- The system shall compute environmental impact metrics based on the user's lifestyle data.
- The system shall generate **five key outputs**:
 - Carbon Emissions (CO₂)
 - Water Footprint
 - Plastic Waste
 - E-Waste Footprint
 - Forest Loss
- The system shall combine these into a **Unified Sustainability Score** for holistic analysis.
- The calculations shall use **scientifically verified datasets** and **standard conversion factors**.

8.4 Prediction Module

- The system shall implement **LSTM/RNN machine learning models** to forecast future environmental impact.
- The prediction shall estimate **5–10 year trends** for each impact type based on user data.
- The module shall display **comparative graphs** showing “current vs future” impact scenarios.
- Users shall be able to simulate different lifestyle changes and see their predicted outcomes.

8.5 User Dashboard

- The system shall display environmental data using **interactive visual dashboards**.
- The system shall include **charts, graphs, and comparisons** between different lifestyle categories.
- The system shall integrate a **3D interactive globe** (using Three.js and Blender models) that visually represents:
 - CO₂ clouds (Travel + Electricity)
 - Drying rivers (Food/Water)

- Plastic waste (Shopping)
- Shrinking forests (Food + Shopping)
- E-waste piles (Gadgets/Tech)
- The visualization shall be dynamic, responsive, and user-interactive.

8.6 Recommendation Module

- The system shall analyze user habits and recommend **sustainable alternatives**.
- Example recommendations:
 - Car → Public Transport
 - Beef → Lentils
 - Fast Fashion → Sustainable Clothing
 - Replace old electronics → Energy-efficient devices
- The system shall generate **personalized suggestions** based on prediction results and user context.
- The recommendation engine shall balance **practicality** and **sustainability** to ensure realistic guidance.

8.7 Multilingual Interface

- The system shall support at least **two languages**: English and Urdu.
- The system shall allow users to **switch language preferences** dynamically.

8.8 Data Storage & Analytics

- The system shall store all user data, calculated metrics, and predictions in **MongoDB[2]**.
- The database shall maintain user **history** to track changes in sustainability over time.
- The system shall allow users to **view trends** and compare past vs present data.
- Data shall be anonymized and stored securely for privacy protection.

9. Non-Functional Requirements

9.1 Performance

- The system shall perform all calculations and display results within **3–5 seconds**.
- The system shall support **real-time rendering** of 3D visualizations without noticeable lag.
- The backend shall efficiently handle multiple concurrent requests using **asynchronous APIs**.

9.2 Accuracy

- The AI/ML models shall be trained and validated on **verified global datasets** (UN, FAO, IPCC, Global Footprint Network[16][17][18][19][20][21]).
- Prediction accuracy shall maintain a **minimum confidence level of 90%**[25][26][30][31] for trend estimation.
- Conversion factors and impact coefficients shall follow **scientifically accepted standards**.

9.3 Scalability

- The architecture shall support **horizontal scaling** to accommodate a large number of users and organizations.
- Cloud hosting (AWS, Vercel, or Heroku) shall ensure automatic scaling during high traffic.
- The database shall be optimized for **big data handling** and **AI model integration**.

9.4 Usability

- The interface shall be **intuitive, minimalistic, and user-friendly** for both technical and non-technical users.
- The system shall implement **responsive design** for all devices (desktop, tablet, mobile).
- It shall include **help tooltips**, guided tours, and simplified dashboards for first-time users.

9.5 Security

- All user data shall be **encrypted** in transit (HTTPS/SSL) and at rest.
- The system shall implement **JWT-based authentication**[24] for secure session handling.
- Sensitive data shall not be exposed or shared without consent.
- The system shall be protected from **common web vulnerabilities** (SQL injection, XSS, CSRF).

9.6 Maintainability

- The codebase shall follow **modular architecture** with clear separation between frontend, backend, and ML layers.
- The system shall use **Git/GitHub**[22] for version control and collaboration.
- Documentation shall be maintained for APIs, modules, and deployment procedures.

9.7 Compatibility

- The web app shall be compatible with all modern browsers (Chrome, Edge, Firefox, Safari).
- The 3D visualization module shall support **WebGL-enabled browsers**[23].
- The application shall function properly on **Windows, macOS, and Android/iOS devices**.

9.8 Response Time

- System response time for data submission, calculation, and visualization shall not exceed **5 seconds** under normal load.
- Page transitions and dashboard updates shall occur **smoothly without reloading** (using React.js).

10. Conclusion

The **Global Lifestyle Impact Predictor (GLIP)** represents a pioneering step toward integrating artificial intelligence, environmental science, and 3D visualization into a unified sustainability analysis platform. Unlike traditional carbon calculators, GLIP provides a **multi-dimensional evaluation** of daily lifestyle habits by calculating five major impact areas—carbon emissions, water footprint, plastic waste, e-waste generation, and forest loss. Through its AI-driven predictive modeling, users gain insights not only into their current environmental footprint but also into **future projections** of their lifestyle impacts.

Moreover, the system’s **personalized recommendation engine** bridges the gap between awareness and action, offering practical, achievable solutions tailored to each individual’s behavior patterns. Its **3D interactive visualization** transforms raw data into engaging, intuitive, and meaningful visuals, promoting environmental learning and sustained behavioral change.

By combining scientific accuracy with accessibility and innovation, GLIP supports both individual and collective progress toward the **United Nations Sustainable Development Goals (SDGs)**, particularly in climate action, responsible consumption, and ecosystem preservation. The project will serve as an educational and analytical tool for users, organizations, and policymakers, ultimately fostering a **global culture of sustainability through technology**.

In essence, GLIP is not just a software system—it is a vision for empowering humanity to understand, predict, and improve its environmental impact through intelligent, data-driven, and visually immersive solutions.

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